

IMAGE CLASSIFICATION USING TRANSFER LEARNING

Internship Project Report

Submitted By

Name: Saikumar B

Course: Bachelor of Technology (Computer Science Engineering)

Year: Second Year

Submitted To

InternSpark Internship Program

Project Details

Project Title: Image Classification using Transfer Learning with ResNet18 on CIFAR-10 Dataset

Deep Learning Model Used:

- ResNet18 (Pretrained)
- Transfer Learning

Dataset Used

- CIFAR-10 Dataset

Techniques Applied

- Data Augmentation
- Transfer Learning
- Model Evaluation
- Deep Learning Classification

Tools & Technologies

- Python
- Jupyter Notebook
- PyTorch
- Torchvision
- Matplotlib
- Scikit-Learn

Performance Metrics

- Accuracy: 80%
- Precision: 81%
- Recall: 80%
- F1-Score: 80%

Submission Date

June 2026

Task 2: Image Classification using Transfer Learning with ResNet18

Objective

The objective of this project is to build an image classification model using deep learning and transfer learning techniques. A pretrained ResNet18 model was used to classify images from the CIFAR-10 dataset into ten categories.

Dataset

The CIFAR-10 dataset contains 60,000 color images of size 32×32 pixels belonging to 10 different classes:

- Airplane
- Automobile
- Bird
- Cat
- Deer
- Dog
- Frog
- Horse
- Ship
- Truck

The dataset is divided into training and testing sets.

Methodology

Data Preprocessing and Augmentation

To improve model generalization, data augmentation techniques were applied:

- Random Horizontal Flip
- Random Rotation
- Tensor Conversion

Model Selection

A pretrained ResNet18 model from PyTorch was used. Transfer learning was applied by replacing the final fully connected layer to classify the 10 CIFAR-10 classes.

Training

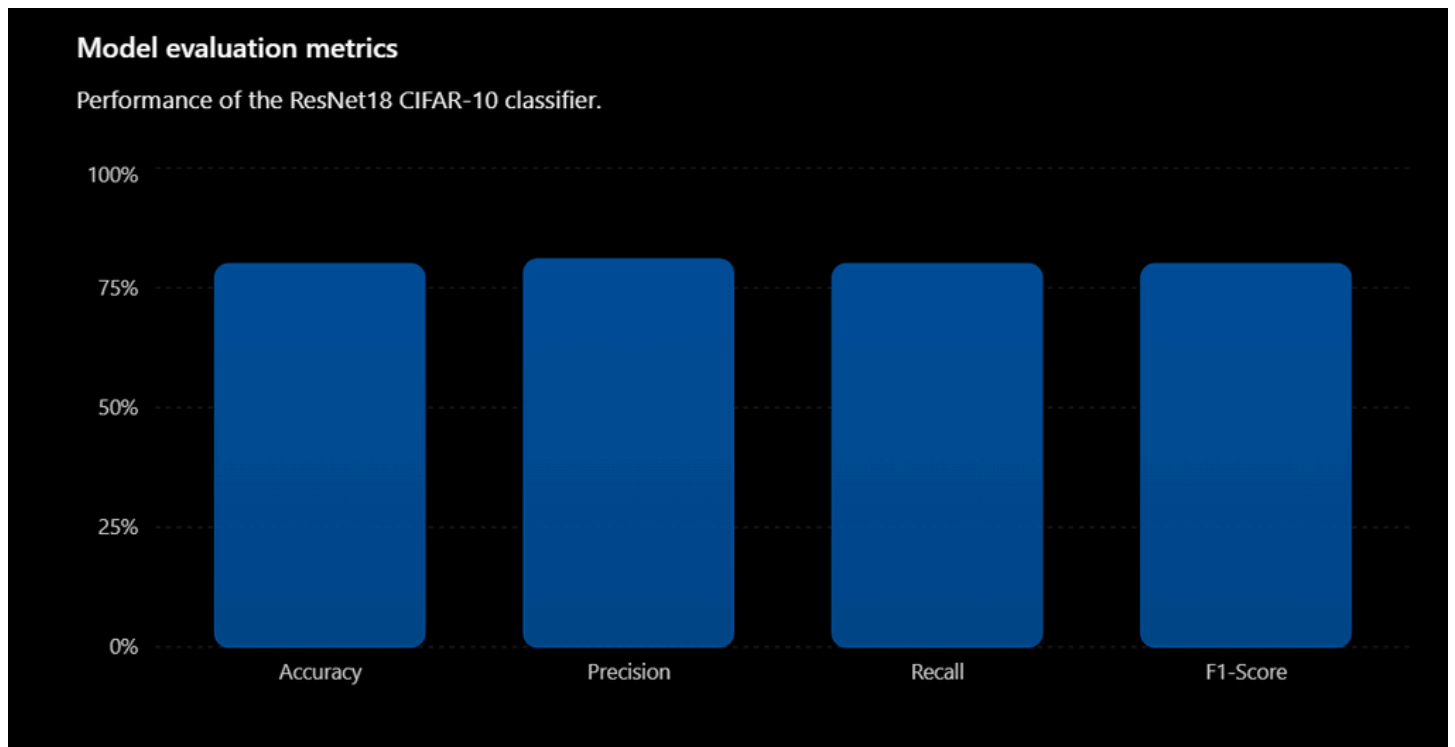
The model was trained using:

- Loss Function: CrossEntropyLoss
- Optimizer: Adam
- Framework: PyTorch

Evaluation Metrics

The model performance was evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score



Results

The trained model achieved the following performance on the test dataset:

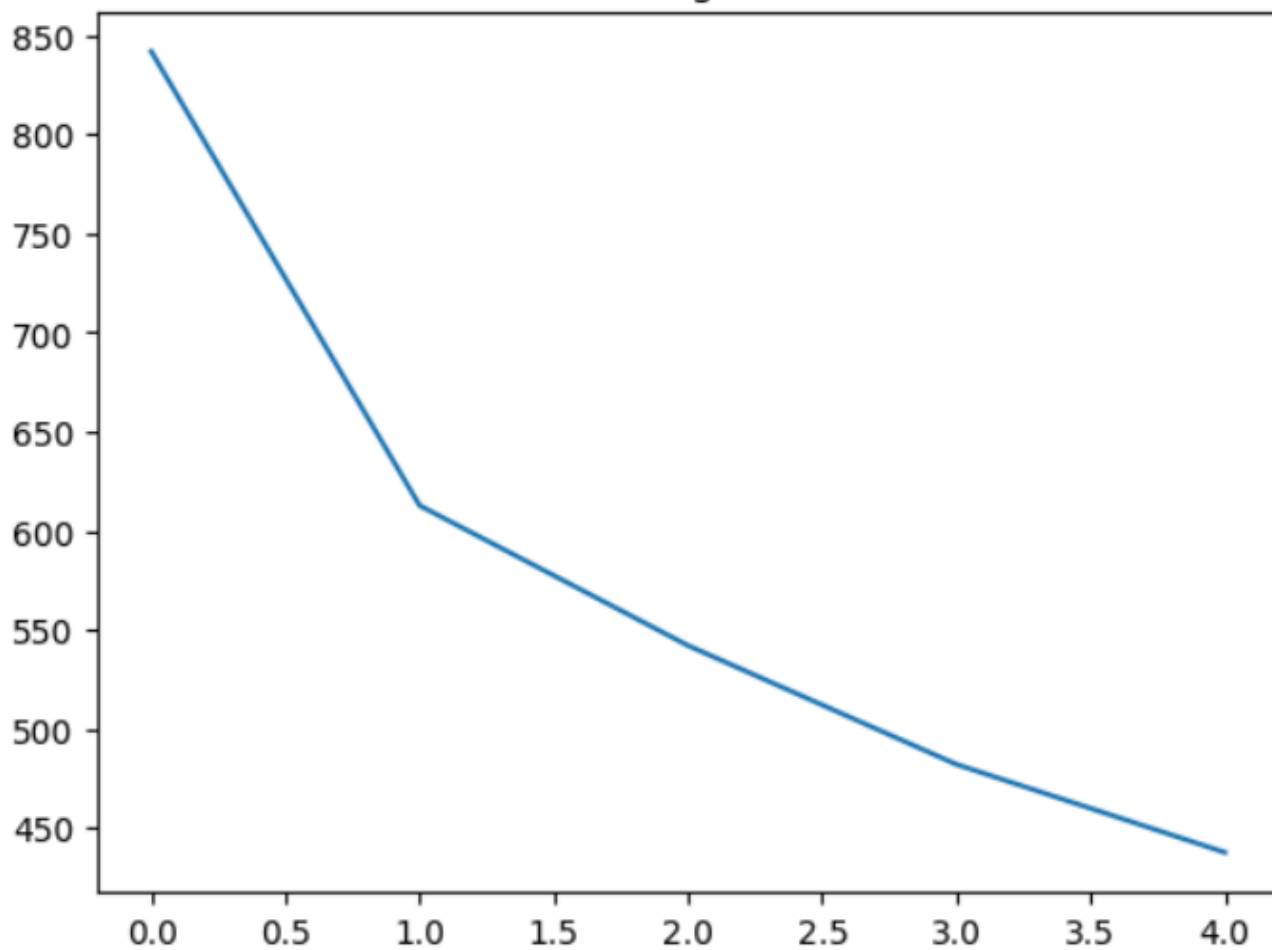
- Accuracy: 80%

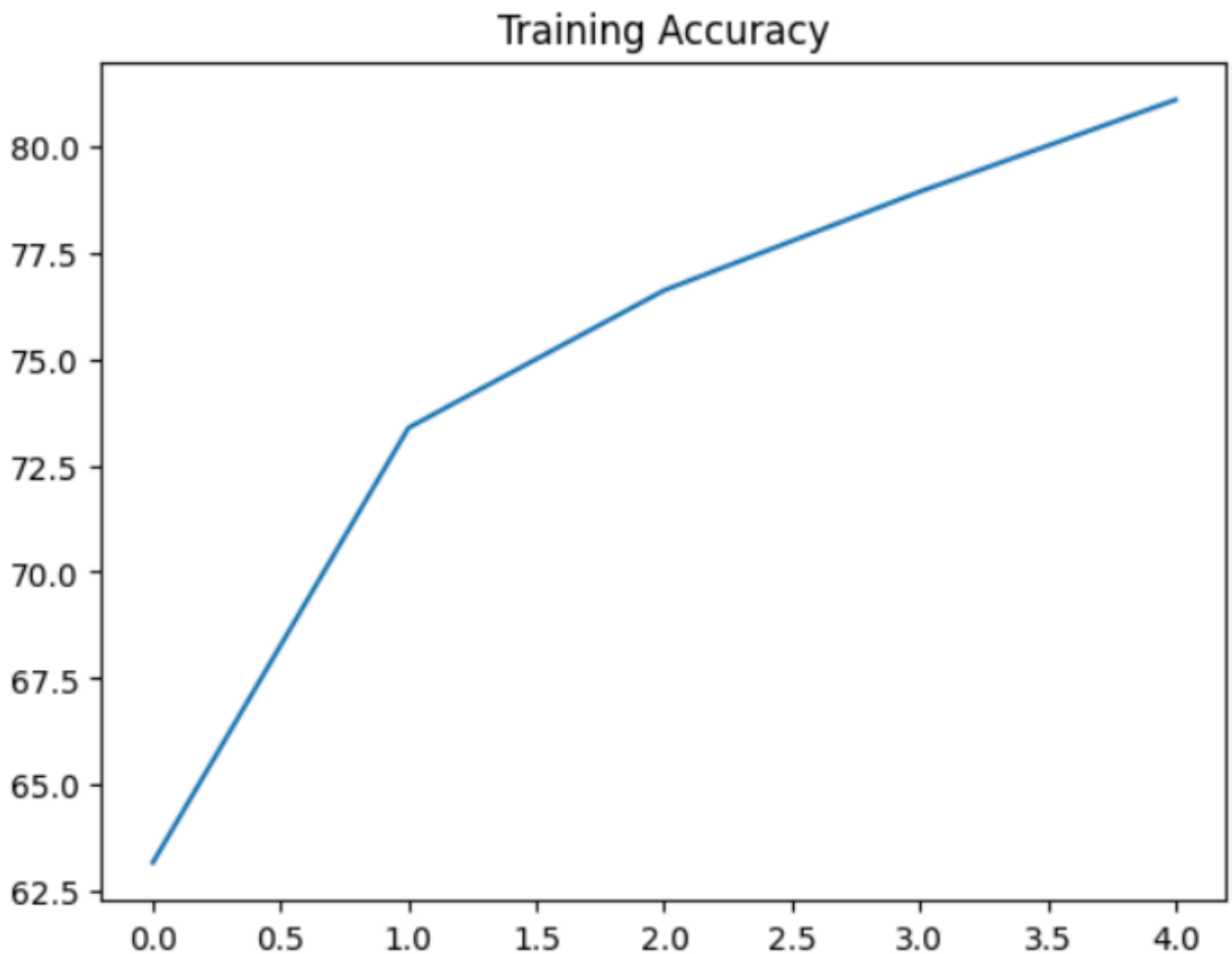
- Precision: 81%
- Recall: 80%
- F1-Score: 80%

	precision	recall	f1-score	support
0	0.83	0.80	0.81	1000
1	0.92	0.86	0.89	1000
2	0.68	0.82	0.74	1000
3	0.62	0.68	0.65	1000
4	0.85	0.79	0.82	1000
5	0.82	0.58	0.68	1000
6	0.90	0.79	0.84	1000
7	0.84	0.87	0.85	1000
8	0.85	0.92	0.88	1000
9	0.78	0.92	0.84	1000
accuracy			0.80	10000
macro avg	0.81	0.80	0.80	10000
weighted avg	0.81	0.80	0.80	10000

Training loss and accuracy curves were generated to visualize the learning process.

Training Loss





Saved Model

The trained model was saved as:

`saved_model.pth`

```
[16]: torch.save(model.state_dict(), "saved_model.pth")  
      print("Model saved successfully!")  
  
Model saved successfully!
```

This file can be loaded later for inference without retraining the model.

Conclusion

This project successfully implemented an image classification system using transfer learning with ResNet18. Data augmentation techniques improved model robustness, and the pretrained architecture enabled efficient learning. The model achieved approximately 80% accuracy on the CIFAR-10 dataset, demonstrating the effectiveness of transfer learning for image classification tasks.

